

Sanpoom Punapanont

Bangkok, Thailand

Mobile: +66805211004, **Email:** p.sanpoom@gmail.com

Personal Details

Name Sanpoom Punapanont

Date and Place of Birth 6 July 1999

Nationality Thai

School and University Education

2018 – 2022 Bachelor's Degree in Robotics and Automation Engineering

Institute of Fields Robotics (FIBO), King Mongkut's University of Technology Thonburi (KMUTT)

GPA: 3.11

2015 – 2018 High school at Pre-Engineering School, King Mongkut's University of Technology North Bangkok

Research Interests

Physical Intelligence, Compliant and Variable-Stiffness Mechanisms, Embodied Intelligence, Robot-Environment Interaction

Experiences

2023–Present Research Engineer at Vidyasirimedhi Institute of Science and Technology

- Conduct robotics research in collaboration with industrial partners.
- Design and experimentally validate variable-stiffness robotic mechanisms.
- Model and simulate variable-stiffness structures and their mechanical behavior.

2021-2023 Engineer at Seagate Technology

- Designed mechanical components and precision equipment for automated manufacturing processes.
- Supported equipment development and functional testing for production lines.

Key publications

Punapanont, Sanpoom, et al. "A Multi-Axis Compliant Helical Joint for Embodied Adaptable Stiffness and Environment-Driven Interaction". **Manuscript under review** at IEEE Transactions on Robotics (T-RO). Submitted June 2026.

- Developed a multi-axis variable-stiffness compliant mechanism using helical structure.
- Demonstrated environment-driven asymmetric propulsion for underwater locomotion with simple sinusoidal actuation.

Punapanont, Sanpoom, et al. "LITHE-joint: Variable Stiffness Compliant Spherical Contact Joint in an Under-Actuated System." 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2025.

- Developed a variable-stiffness compliant spherical joint for quadruped locomotion.
- Demonstrated directional locomotion through asymmetric body stiffness without changing the basic actuation pattern.

Punapanont, Sanpoom, et al. "REFINE-bot: Furnace Cleaning Robot for Heat-transfer Efficiency Improvement." 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2025.

Romtrairat, P., et al. "Development of 7-degree-of-freedom passive manipulator for arthroscopy." *AIP Conference Proceedings*. Vol. 3086. No. 1. AIP Publishing LLC, 2024.

Technical Skills

Hardware	CAD design e.g. SolidWorks, Fusion 360
	Compliant mechanism design
	Pneumatic systems
Modeling & Computation	MATLAB
	Python
	Kinematic and dynamic modeling

Languages

Thai	Native
English	B2 - IELTS overall 6.5 (no sub band less than 6)

Honors/ Awards

2025	VISTEC honors for outstanding achievements.
2019-2021	KMUTT Full Scholarship for Creativity and Innovation.